

JAY DARJI

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Machine Learning Engineer with ~3 years of experience building and deploying end to end machine learning systems. Specialized in model development experimentation and data centric learning using Python, PyTorch and AWS. Experienced in transforming peer-reviewed research findings into robust production ML solutions.

TECHNICAL SKILLS

- Python, Java, SQL, MATLAB, Machine Learning, NLP, LLMs, RAG, Computer Vision, Docker, AWS, Time-Series, Signal Processing, PyTorch, TensorFlow, Scikit-learn, Transformers, GitHub, Flask, Tableau

EDUCATION

San Jose State University, San Jose

GPA: 3.7/4

Master of Science in Electrical Engineering (concentration in Artificial Intelligence)

Aug 2024 - Dec 2025

- Relevant Courses:** Machine Learning, Neural Network, Computer Vision, Reinforcement Learning, AI for DSP

Institute of Infrastructure Technology Research and Management (IITRAM), Ahmedabad

GPA: 3.2/4

Bachelor of Technology in Electrical Engineering

Aug 2017 - May 2021

EXPERIENCE

Machine Learning Research Assistant

Jun 2025 - present

San Jose State University, San Jose

- Developed an **XAI** system using **DINOv3 ViT**, achieving **83.25%** accuracy on 4-class brain tumor classification.
- Implemented **Occlusion**, Deepdream and Lucent visualizations to interpret a **256D** classifier **feature space**.
- Built a PyTorch & **GPT-4o-mini** pipeline to synthesize visual XAI data into natural language clinical **reports**.
- Architected a **multi-agent** framework for automated feature extraction, model training, and report synthesis.

Data Scientist

Sep 2022 - Jun 2024

Rhenix Lifesciences LLP, Hyderabad

- Built **hybrid CNN-LSTM model** for **CGM forecasting**, achieving **10 mg/dL RMSE** for 15-minute horizons.
- Outperformed published benchmarks with a **17 mg/dL RMSE** for 60-minute forecasts using time-series data.
- Built **Random Forest model** for **HRV-based sleep classification**, reaching **93% accuracy and 0.96 AUC**.
- Designed novel binning-based imputation method**, improving **RMSE in 70-90%** of **time-series** cases.

Full Stack Software Developer

Dec 2021- Apr 2022

Cognizant Technology Solutions Pvt. Ltd., Bengaluru

- Developed and maintained **Java Spring Boot applications**, enhancing system performance and scalability.
- Designed and **integrated REST APIs**, ensuring efficient front-end and backend communication.
- Optimized MySQL queries**, cutting execution time **by 20%** through indexing and query tuning.

Machine Learning Intern

Jun 2019 - Aug 2019

Institute of Infrastructure Technology Research and Management, Ahmedabad

- Trained an **ensemble bagged trees** model for classifying sleep disorders, attaining an accuracy of **95%**.
- Mentored 10 students** on experimental design and ML techniques; **co-authored** peer-reviewed research.

ACADEMIC PROJECTS

End to End NLP Text Summarization Pipeline Deployed on AWS with Flask

- Built an **NLP** pipeline using Hugging Face **Transformers** for extractive and abstractive text summarization.
- Automated preprocessing, model training, and **deployment** workflow with **Docker** and **AWS EC2**.
- Integrated monitoring and **logging** for real time summarization through a **Flask** based web interface.

NLP-Powered Resume Matching App Using Embedding Models and Gemini LLM

- Developed a web app to match resumes with job descriptions using SBERT, **OpenAI**, and Gemini embeddings.
- Ranked candidates by **cosine similarity** scores with **LLM-generated** fit summaries and verdicts.
- Integrated a **fine-tuned SBERT** model and modular backend for real-time analysis via **Streamlit**.

RESEARCH PUBLICATIONS

- Automated identification of sleep disorders using wavelet-based features extracted from electrooculogram and electromyogram signals **DOI:** [10.1016/j.comptbiomed.2022.105224](https://doi.org/10.1016/j.comptbiomed.2022.105224)
- Handling missing data in the **time-series** data from wearables **DOI:** [10.5772/intechopen.1002536](https://doi.org/10.5772/intechopen.1002536)
- Efficient use of binned data for **imputing** uni-variate time-series data **DOI:** [10.3389/fdata.2024.1422650](https://doi.org/10.3389/fdata.2024.1422650)